FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Advanced)-2021

PAPER –1

TEST DATE-30-05-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

  Section-A (01 – 06, 19 – 24, 37– 42) contains **18 multiple** choice questions which have **ONLY ONE CORRECT ANSWER**. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

  Section-A (07 – 12, 25 – 30, 43 – 48) *this section contains* ***18 multiple*** *choice questions.*
  Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
  For each question, choose the option(s) corresponding to (all) the correct answer(s)
  Answer to each question will be evaluated according to the following marking scheme:

  Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
  Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
  Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
  Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
  Negative Marks **: –2** In all other cases.

  Section-C (13 – 18, 31– 36, 49 – 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

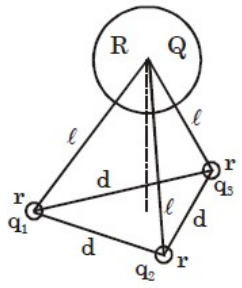
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. Three identical uncharged metal spheres are at the vertices of an equilateral triangle. One at a time, a small sphere is connected by a conducting wire with a large metal sphere that is charged. The center of the large sphere is in the straight line perpendicular to the plane of equilateral triangle and passing through its centre (see figure). As a result, the first small sphere acquires charge q_1 and second charge $q_2\,(q_2 < q_1)$.



The charge that the third sphere q_3 will acquire is:
$(\text{Assume } \ell >> R, \ell >> r, d >> R,\ d >> r)$

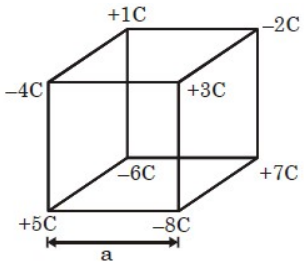
(A) $\dfrac{q_1^2}{q_2}$

(B) $\dfrac{q_2^2}{q_1}$

(C) $\sqrt{q_1 q_2}$

(D) $\dfrac{q_1 + q_2}{2}$

2. The electric flux passing through the cube for the given arrangement of charges placed at the corners of the cube (as shown in the figure) is:



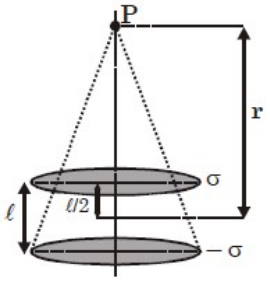
(A) $\phi = \dfrac{1}{2\in_0}$

(B) $\phi = \dfrac{-1}{2\in_0}$

(C) $\phi = \dfrac{-1}{\in_0}$

(D) $\phi = \dfrac{1}{\in_0}$

3. Two parallel disks each having radius R are separated by a distance ℓ. The surface charge densities are σ and $-\sigma$. The electric field at point P, a large distance r along the axis of the disks is:



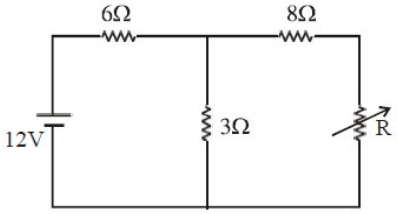
(A) $\dfrac{\sigma R^2 \ell}{\varepsilon_0 r^3}$

(B) $\dfrac{2\sigma R^2 \ell}{\varepsilon_0 r^3}$

(C) $\dfrac{\sigma R^2 \ell}{4\varepsilon_0 r^3}$

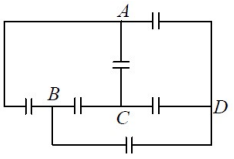
(D) $\dfrac{\sigma R^2 \ell}{2\varepsilon_0 r^3}$

4. The maximum power delivered to resistance R is:



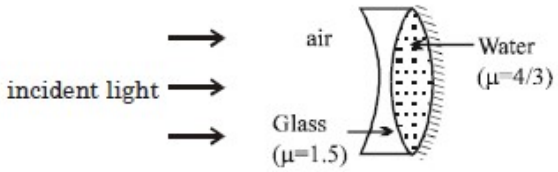
(A) 0.2W
(B) 0.4W
(C) 0.8W
(D) 0.16W

5. Choose the correct statement for the given capacitor arrangement. (All the 6 capacitors are of same capacitance).



(A) If a battery is connected across AB, all 6 capacitors get charged.
(B) If a battery is connected across AC, all 6 capacitors get charged.
(C) If a battery is connected across AD, all 6 capacitors get charged.
(D) It is not possible to charge all the 6 capacitors using single source.

6. The radius of curvature of the left and right surface of the concave lens are 10cm and 15 cm respectively. The radius of curvature of the mirror is 15 cm. (Consider direction of incident light as positive)

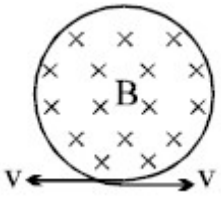


(A) Equivalent focal length of the combination is $-36cm$
(B) Equivalent focal length of the combination is $+36cm$
(C) The system behaves like a concave mirror.
(D) The system behaves like a convex mirror.

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

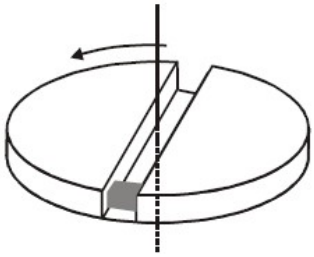
7. A circular conducting loop of radius r_0 and having resistance per unit length λ as shown in the figure is placed in a magnetic field B which is constant in space and time. The ends of the loop are crossed and pulled in opposite directions with a velocity v such that the loop always remains circular and the radius of the loop goes on decreasing, then



(A) Radius of the loop changes with $r = r_0 - vt/\pi$
(B) EMF induced in the loop as a function of time is $e = 2Bv\left[r_0 - vt/\pi\right]$
(C) Current induced in the loop is $I = \dfrac{Bv}{2\pi\lambda}$

(D) Current induced in the loop is $I=\frac{Bv}{\pi\lambda}$

8. A large, horizontal disk of radius R, shown below, starts to rotate from rest with an angular acceleration of α. The rotation is about a vertical axis through the centre of disk. The disk contains a narrow channel of length 2R and rectangular cross-section. Gravity acts in the vertical direction with an acceleration of g. There is a small rectangular puck that just fits easily in the aforementioned channel, as shown. The puck is situated a distance r from the axis of rotation.



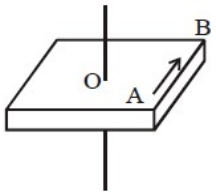
(A) If the sides of the channel are frictionless but the bottom of the channel has a static coefficient of friction μ, then puck begins to slide at $t=\sqrt{\frac{\mu}{\alpha}}$

(B) In case mentioned in option (A), $t=\sqrt{\frac{\mu g}{r\alpha^2}}$

(C) Now, instead, the situation is that the bottom of the channel is frictionless but the walls have a static coefficient of friction μ. Now puck begins to slide at $t=\sqrt{\frac{\mu g}{r\alpha^2}}$

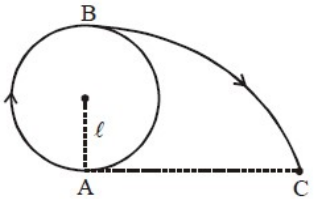
(D) In case mentioned in option (C), $t=\sqrt{\frac{\mu}{\alpha}}$

9. A horizontal square platform of mass m and side a is free to rotate about a vertical axis passing through its centre O. The platform is stationary and a person of the same mass (m) as the platform is standing on it at point A. The person now starts walking along the edge from A to B (see figure). The speed v of the person with respect to the platform is constant. Taking v = 5 m/s and a = 1m.



(A) Angular velocity of platform when the person is at A is 6 rad/s.
(B) Angular velocity of platform increases first and then decreases again as it reaches B.
(C) Angular velocity of platform decreases first and then increases again as it reaches B.
(D) Angular velocity of platform remains constant as the person goes from A to B.

10. The body is rotated in the vertical plane by means of a thread of length ℓ with minimum possible speed. At the highest point B the thread breaks and the body moves under gravity and strikes at C. Then:



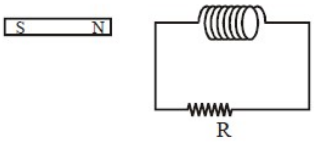
(A) The horizontal range AC is 2ℓ

(B) The horizontal range AC is $\sqrt{2}\ell$

(C) Tangential acceleration of the body at point C is $\frac{2}{\sqrt{5}}g$

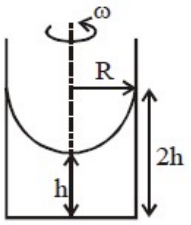
(D) Tangential acceleration of the body at point C will be $\frac{1}{\sqrt{5}}g$

11. Here inductor is connected to a resistor as shown. When a magnet is moved towards the coil with a constant speed till it comes out of coil completely. As seen by observer looking at the rectangular loop as shown.



(A) Current first flows clockwise and then anticlockwise in loop.
(B) Current first flows anticlockwise and then clockwise in loop.
(C) Magnet first experiences a repulsive force and then an attractive force.
(D) When rate of change of flux due to magnet is $\frac{d\phi}{dt}$, current in coil is $\frac{d\phi}{Rdt}$

12. A fluid of density ρ is rotated with a certain angular velocity in a cylindrical vessel as shown:



(A) The angular velocity is $\omega=\sqrt{\frac{2gh}{R^2}}$
(B) The angular velocity is $\omega=\sqrt{\frac{4gh}{R^2}}$
(C) The kinetic energy of fluid is $\frac{16\pi\rho gh^2R^2}{15}$
(D) The kinetic energy of fluid is $\frac{5\pi\rho gh^2R^2}{6}$

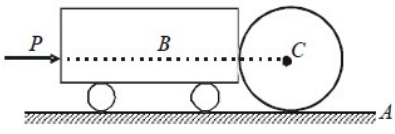
SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A point like sound source emits sound in all direction uniformly. An observer who is at a distance of $50\ \text{m}$ from the source detects sound of intensity $10^{-8}\ \text{watt/m}^2$. If the bulk modulus of air is $1.6\times10^5\ \text{N/m}^2$ and velocity of sound is 340 m/s. Find pressure amplitude at the position of observer. If your answer is $x\times10^{-3}\ \text{N/m}^2$ fill value of [x]_____

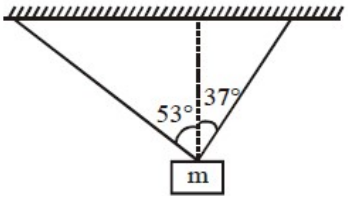
14. When the voltage applied to an X-ray tube increased from $V_1=15.5\,\text{KV}$ to $V_2=31\text{KV}$, the wavelength interval between the K_a line and the short wavelength cut-off of the continuous X-ray spectrum increases by a factor of 1.3. Find the atomic number of the element of the target. If your answer is N fill value of $\frac{N}{13}$. (Take: he = 1240 eV and R = 1×10^7/m)

15. A $10\,\text{kg}$ block B rests on smooth rollers of negligible mass and is in contact with a $2.0\,\text{kg}$ solid cylinder of the radius 0.1m as shown. The rollers ensure that there is no friction between B and the ground A below B. It is know that $\mu_k=\mu_s=0.5$ between B and C as well as between C and ground A. Find the magnitude of the horizontal force P (in N) for which the cylinder will roll with an angular acceleration of $20\,\text{rad/s}^2$ on ground A without slipping. Fill $\frac{P}{16}$ on OMR sheet.

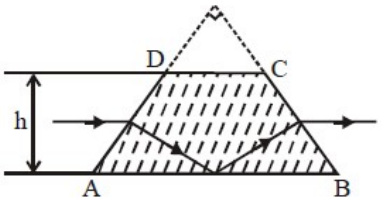


16. A vernier calliper has no zero error, one main scale division = 1 mm and 9 main scale division = 10 vernier scale divisions. A spring is held between jaws without exerting force on jaws. Main scale reading is 3.4cm and 5th vernier scale division coincides with a main scale division. When jaws are pressed by force of 10 N on both sides, main scale reading remains 3.4cm, but 1st division of vernier scale coincides with a main scale division. Now spring is removed, cut into two pieces and when inserted between jaws without any force, main scale reading $=1.1\text{cm}$ and 5th vernier scale division coincides with a main scale. If we apply force of 30 N on both the jaws, main scale reading remains 1.1cm and x vernier scale division coincides with a main scale division. Fill x in OMR sheet.

17. A heavy mass is hung from ceiling as shown. Both wires are identical. A pulse is produced in both the strings simultaneously at the point of contact with mass. If it reaches the ceiling in $3\sqrt{3}\times10^{-3}$ sec for string on the right, in what time will it reach the ceiling for the left string (in 10^{-3} sec).



18. For the inversion of the image we often use a Dove prism (see figure), representing truncated regular rectangular isosceles prism. Find approximately the length of side AB (in cm) at which the beam of light falling on side AD will come out of side BC completely. The height of the trapezium ADCB is equal to $h=1.05\,\text{cm}$. Refractive index of the prism is 1.41. Angle A and B are $45°$ each. (Given: $\sqrt{2}=1.41$, $\sqrt{3}=1.73$)



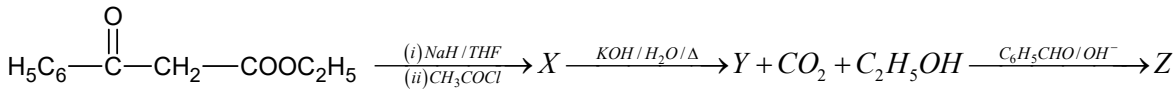
Chemistry

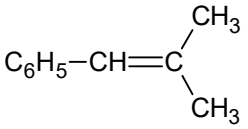
PART – II

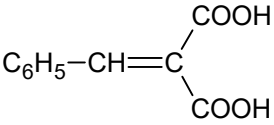
SECTION – A
(One Options Correct Type)

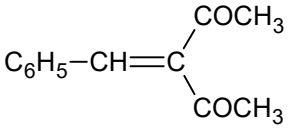
This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. Identify the final product $'Z'$



(A) 

(B) 

(C) 

(D) $C_6H_5-CH=CH-COCH_3$

20. A chromatography column, packed with silica gel as stationary phase, was used to separate a mixture of compounds consisting of (A) benzanilide (B) aniline and (C) acetophenone. When the column is eluted with a mixture of solvents, hexane: ethylacelate (20:80), the sequence of obtained compounds is:
(A) (A), (B) and (C)
(B) (B), (A) and C
(C) (C), (A) and (B)
(D) (B), (C) and A

21. Calculate ΔG° for the following cell reaction

$Zn(s)+Ag_2O(s)+H_2O(I)\rightarrow Zn^{2+}(aq)+2Ag(s)+2OH^-(aq)$

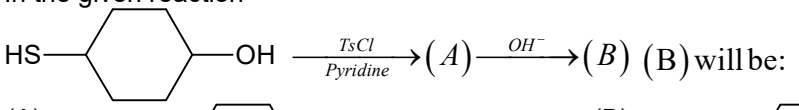
$E^0_{Ag^+/Ag}=+0.80V$ and $E^0_{Zn^{+2}/Zn}=-0.76V,\ F=96500$

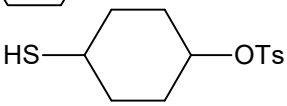
(Given: K_P of $AgOH=2\times10^{-8}$)

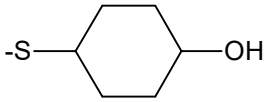
(A) $-305kJ/mol$
(B) $212kJ/mol$
(C) $305kJ/mol$
(D) $301kJ/mol$

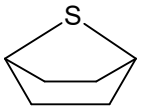
22. The coagulation of 10 cm^3 of gold solution by 1 ml 10% NaCl solution is completely prevented by addition of 0.25 g of starch to it. The gold number of starch is:
(A) 0.025
(B) 0.25
(C) 2.5
(D) 250

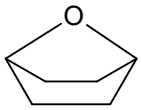
23. In the given reaction



(A) 

(B) 

(C) 

(D) 

24. A red solid is insoluble in water. However it becomes soluble if some KI is added to water. Heating the red solid in a test tube results in liberation of some violet coloured fumes and droplets of a metal appear on the cooler parts of the test tube. The red solid is:

(A) HgI_2

(B) HgO

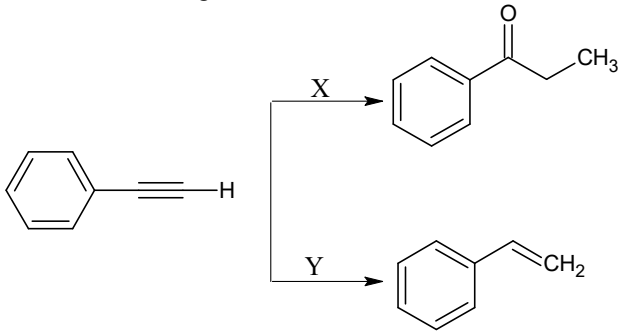
(C) Pb_3O_4

(D) $(NH_4)_2Cr_2O_7$

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

25. Consider following conversions



Where X and Y are reagent or set of reagents. Which of the following statement(s) is/are correct?

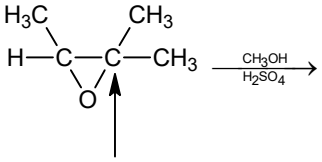
(A) $Na/liq.NH_3$ can be used as Y

(B) $H_2/Pd.BaSO_4$ can be used as Y

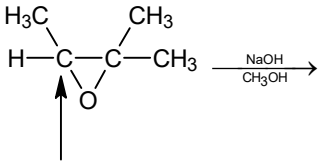
(C) $NaNH_2/Me\text{-}1/1\%\,HgSO_4$ can be used as X

(D) $KOH/Me\text{-}1/1\%\,HgSO_4$, dil H_2SO_4 can be used as X

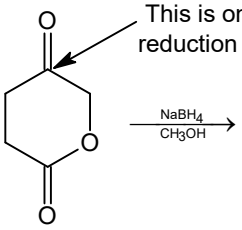
26. Which is/are correct statements?

(A) 

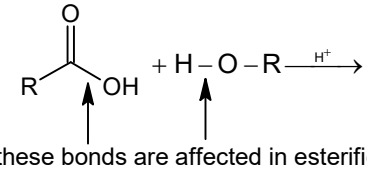
nucleophile attacks here when epoxy linkage is cleaved

(B) 

nucleophile attacks here

(C) 

This is only affected in reduction to 2° alcohol:

(D) 

these bonds are affected in esterification

27. $3HCHO + CH_3CHO \xrightarrow{NaOH} A$

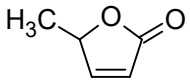
A formed can:

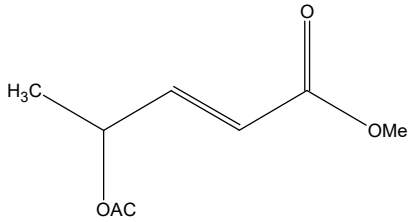
(A) reduce Tollen's reagent
(B) give Cannizzaro reaction
(C) react with Na
(D) give green colour with $Cr_2O_7^{2-}/H^+$

28. Which statements(s) is/ are correct

(A) Among transition metals only Zn, Cd, and Hg have one or more typical metallic structures at normal temperatures.
(B) Among transition metals only V, Nb, Ta, Mo and W have bcc structures and all other can have different Cr, Mo and W. (in their respective series).
(C) maximum m.pt in transition metal (3d series) series are of Cr, Mo and W. (in their respective series)
(D) The m.pt of corresponding elements generally follow order 3d series < 4d series < 5d series

29. Find which is/are correct.

(A) Amides R_2NCOR are much more nucleophilic on O than they are on N.
(B) Esters are much less electrophilic at C than Ketones.
(C) Acyl chlorides $RCOCl$ are much more acidic than esters
(D) 

is more acidic than 

30. One mole triatomic vapours of an unknown substance effuses 4/3 times faster than 1 mole O_2 under same conditions. If the density of unknown vapours at pressure P and temperature T is d, which of the following holds true for the unknown substance?

(A) $d_{N.T.P.} = 0.8035\, g/l$
(B) $Z(\text{atomic number}) = 6$
(C) $Z(\text{compress ibility factor}) = 18P/dRT$
(D) Vapour density=9

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. In the following compound some undergo disproportionation with conc. NaOH and some do not give disproportionation with conc. NaOH:
$F_2, Cl_2, Br_2, N_2, CH_3COCH_3, CCl_3CHO, (C_6H_5)_3CCHO, P_4, I_2, (CH_3)_2CHCHO, HCHO.$
'x' are the number of compound which undergo disproportionation and 'y' are the number of compound which do not undergo disproportionation then x/y is:

32. The standard Gibbs free energy change of the reaction shown below is $-2.7\,kJ.mol^{-1}$.
$Sn(s) + Pb^{2+} = Sn^{2+} + Pb(s)$
Given that $E^0(Pb^{2+}/Pb) = -0.126V$, Find the value of magnitude of $E^0(Sn^{2+}/Sn)$

33. Calculate the molar solubility of $AgBr$ in 0.1 M $Na_2S_2O_3$ (aq.) solution. K_{sp} of $AgBr$ $=5\times10^{-13}$ and $K_f\left[Ag(S_2O_3)_2^{3-}\right] = 5\times10^{13}$. Give S if molar solubility is $S\times10^{-1}$ _____

34. Compute the percentage void space per unit volume of unit cell in zinc sulphide structure.

35. 10g of NH_4Cl (mol. Weight = 53.5) when dissolved in 1000 g of water lowered the freezing point by $0.637°C$. Calculate the degree of hydrolysis of the salt if its degree of dissociation is 0.75. The molal depression constant of water is $1.86\,kg\,mol^{-1}K$.
ROUND OFF YOUR ANSWER CORRECT UPTO 2 DECIMAL PLACES

36. The vapour pressure of a certain liquid is given by the equation:
$\log_{10}P = 3.54595 - \frac{313.7}{T} + 1.40655\log_{10}T$ where P is the vapour pressure in mm and $T = Kelvin$ temperature. Determine the molar latent heat of vaporisation as a function of temperature. Calculate its value at $80K$.

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. Consider a real valued continuous function f such that $\mathrm{f}(x)=\sin x+\int\limits_{-\pi/2}^{\pi/2}\left(\sin x+t\,f(t)\right)dt$. If maximum value of the function is equal to $k(\pi+1), k\in N$, then the value of k is:

(A) 1
(B) 2
(C) 3
(D) 4

38. $\int\limits_{2\pi/3}^{\pi}\sqrt{(\sin x+\sin 2x+\sin 3x)^2+(\cos x+\cos 2x+\cos 3x)^2}\,dx$ equals:

(A) $\left(\sqrt{3}-\frac{\pi}{3}\right)$
(B) $\left(\frac{\pi}{3}+\sqrt{3}\right)$
(C) $\frac{5\pi}{3}-\sqrt{3}$
(D) $\sqrt{3}+\frac{5\pi}{3}$

39. Let $f(x)=\cos^{-1}x+\left|\cos^{-1}x\right|+\cos^{-1}|x|$. If $f(x)=k$ has one distinct solution then range of k is:

(A) $\left[0,\frac{\pi}{2}\right]$
(B) $[0,\pi]$
(C) $\left[0,\frac{3\pi}{2}\right]$
(D) $[0,2\pi]$

40. Let $T_n=3n-2$ and $T'_m=7m+2$ be two sequences each having 2004 terms. The number of common terms to the two sequences is:

(A) 283
(B) 284
(C) 285
(D) 286

41. In an equilateral triangle R : r : r_1 is equal to
(A) 1 : 1 : 1
(B) 1 : 2 : 3
(C) 2 : 1 : 3
(D) 3 : 2 : 4

42. If the radius of in-circle and radii of ex-circles of a triangle are consecutive of a geometric progression then the largest angle of the triangle will be:
(A) $60°$
(B) $90°$
(C) $120°$
(D) None of these

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

43. If the cubic equation $x^3 - x + 1 = 0$ has roots α, β, γ, then:
(A) $\alpha^2 + \beta^2 + \gamma^2 = 2$
(B) $\alpha^3 + \beta^3 + \gamma^3 = -3$
(C) $\alpha^4 + \beta^4 + \gamma^4 = 2$
(D) $\alpha^5 + \beta^5 + \gamma^5 = -5$

44. Let $A = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 1 & 2 \\ 3 & 1 & 1 \end{pmatrix}$ and $A^{-1} = \alpha A^2 + \beta A + \gamma I,$ then which of the following holds good?
(A) $\alpha + \beta = 1$
(B) $3(\alpha + \beta + \gamma) = 10$
(C) $8(\beta + \gamma) = 11$
(D) $8(\alpha + \gamma) = 3$

45. $f(x) = \begin{cases} 3x^5 + 2x^3, & x \in \text{Rational} \\ 0, & x \in \text{Irrational} \end{cases}$, then
(A) f(x) is continuous exactly at one point
(B) f(x) is discontinuous $\forall$ x $\in$ R
(C) f(x) is non-differentiable $\forall$ x $\in$ R
(D) f(x) is differentiable exactly at one point

46. If vector $\vec{a},\vec{b},\vec{c}$ are linearly independent and $(\vec{d}\times\vec{b})\times(\vec{c}\times\vec{a})+(\vec{d}\times\vec{c})\times(\vec{a}\times\vec{b})+(\vec{d}\times\vec{a})\times(\vec{b}\times\vec{c}) = \vec{r}$
(A) $\vec{r}$ is collinear with $\vec{d}$
(B) $\vec{d}\cdot(\vec{a}\times\vec{r}) = 0$
(C) If $\vec{c}\cdot\vec{m} = 0, |\vec{m}| = |\vec{c}|, \left[(\vec{b}+\vec{c})\times(\vec{b}+\vec{m})\right]\times(\vec{c}\times\vec{m})\cdot(\vec{c}) = \left[\vec{b}\ \vec{d}\ \vec{c}\right]|\vec{d}|^2$, where $\vec{m} = \dfrac{\vec{r}}{-2\left[\vec{a}\ \vec{b}\ \vec{c}\right]}$
(D) $\vec{r}$ is a linear combination of $\vec{a}$, $\vec{b}$ and $\vec{c}$

47. The value of the integral $\int e^{\sin^2 x}\left(\cos x+\cos^3 x\right)\sin x\,dx$ is

(A) $\frac{1}{2}e^{\sin^2 x}\left(3-\sin^2 x\right)+c$

(B) $e^{\sin^2 x}\left(1+\frac{1}{2}\cos^2 x\right)+c$

(C) $e^{\sin^2 x}\left(3\cos^2 x+2\sin^2 x\right)+c$

(D) $e^{\sin^2 x}\left(2\cos^2 x+3\sin^2 x\right)+c$

48. Let ABC be a triangle with fixed base BC = 1 and vertex 'A' is variable. If one base angle is double of the other base angle then locus of A is a hyperbola
(A) Whose eccentricity is 4
(B) Whose L.R is 4
(C) Whose eccentricity is 2
(D) Whose L.R. is 2.

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. If a, b, c be the sides of a triangle ABC with right angle at C. The medians AD and BE have slopes 1 and 2 respectively. Then ab = kc^2 where 36k equals _____

50. The greatest value of the function $f(x)=2.3^{3x-3^{2x}}.4+2.3^{x}$ in the interval $[-1,1]$ is_____

51. Let n be a positive integer. If the number of integers in the domain of the function $f(x)=\ln\left((1-x)(x-n)\right)$ is $2n-11$, then $\cot^{-1}(\cot n)$ is equal to: $[\text{Use } \pi=3.14]$

52. If α is a real root of the equation $x^3+3x+\tan 2=0$ then the value of $\tan^{-1}\alpha+\tan^{-1}\frac{1}{\alpha}+\frac{\pi}{2}$ is equal to: $[\text{Use } \pi=3.14]$

53. The plane $\frac{x}{1}+\frac{y}{2}+\frac{z}{3}=1$ intersect x-axis, y-axis, z-axis at A, B, C respectively. If the distance between origin and orthocentre of ΔABC is equal to k, then the value of 'k' is_____

54. Let $S=\frac{1^2}{1.3}+\frac{2^2}{3.5}+\frac{3^2}{5.7}+.....+\frac{500^2}{999.1001}$. Then the total number of divisions of $[S]$ is____
[where [.] denotes GIF.]